

Mid-Term Examination

ACOL 216: Introduction to Computer Architecture

IIT Delhi - Abu Dhabi · Semester II, 2025-26

Total Marks: 25

Time: 50 minutes

1. Prove each of the following Boolean expressions using Boolean algebraic laws.

(i) $(A' + B') \cdot (A' + B) \cdot (A + B' \cdot D + C) = A' \cdot B' \cdot D + A' \cdot C$

(ii) $A \cdot B' + A' \cdot B' + A \cdot B' \cdot C \cdot D = B'$

(2.5 × 2 = 5 marks)

2. A number is known as a cubic Armstrong number if the sum of the cubes of its individual decimal digits is equal to the number itself. For example, 153 is a cubic Armstrong number ($153 = 1^3 + 5^3 + 3^3$).

You are given a number in register r_0 , and it is known to be between 1 and 1,000,000. **Write a piece of assembly code in SimpleRisc to determine if this number is a cubic Armstrong number.** Save 1 in register r_1 if it is a cubic Armstrong number; otherwise, save 0. (5 marks)

3. Draw the circuit diagram of a standard 4-bit ripple-carry adder. Then, explain how you can configure or augment this exact same adder circuit (without fundamentally changing its internal structure) to perform the following specific operations on a 4-bit input A :

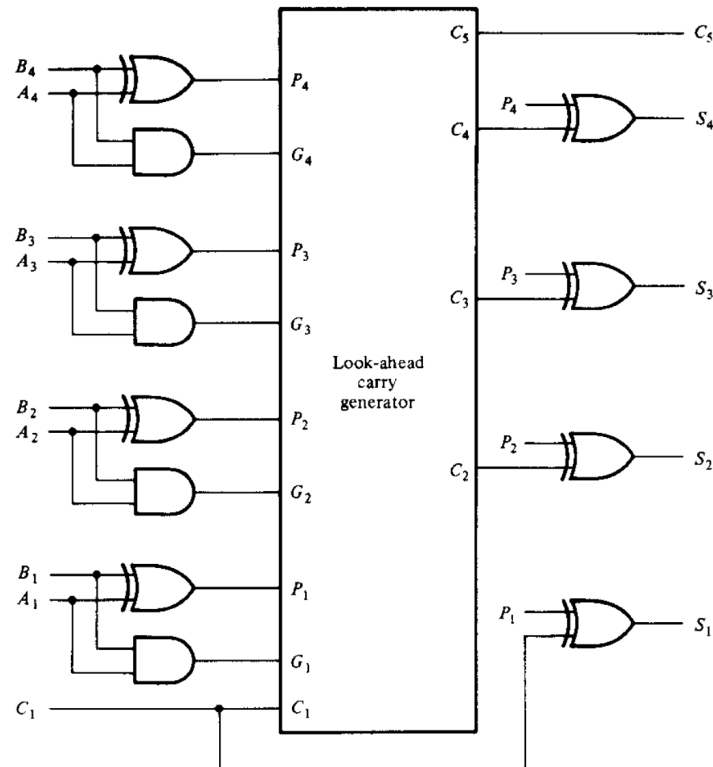
(i) 2's Complement Subtraction: Computing $A - B$ (where B is a second 4-bit input).

(ii) Increment: Computing $A + 1$.

(iii) Decrement: Computing $A - 1$.

(2+1+1+1 = 5 marks)

4. The figure below shows the schematic of a 4-bit look-ahead carry generator. The inputs are $A_1 \dots A_4$ and $B_1 \dots B_4$, with an initial carry-in of C_1 .



Derive the two-level Boolean expression for the final output carry C_5 strictly in terms of the generate variables ($G_1 \dots G_4$), propagate variables ($P_1 \dots P_4$), and the initial carry C_1 . You must show the step-by-step unrolling of the equations to fully expand the final expression. (5 marks)

5. (a) State the fundamental physical components used to store a single bit of information in a 6-transistor SRAM cell versus a DRAM cell.

- (b) Explain why a DRAM array requires *periodic refreshing* while an SRAM array does not. What specific steps occur during a DRAM refresh operation?
- (c) In an array of SRAM cells, reading is optimized using a technique called *pre-charging*. Briefly explain the physical issue this technique solves and how it achieves faster read times.

(1+2+2 = 5 marks)